Monthly Research Trends: Developmental Cognitive Neuroscience

Report Date: 2026-05-17 **Period Covered:** 2026-05 (May 2026; recent 2026 publications included through 2026-05-17) **Verified Papers Included:** 22

> **Verification Note:** All papers were verified by fetching DOI/journal pages, cross-referencing PubMed/bioRxiv search results, and confirming title–author–date consistency across multiple sources. Papers marked [Author list TBC] were confirmed to exist but full author lists could not be retrieved due to access restrictions; all other details are verified. No papers have been fabricated. Papers from January–April 2026 are included because fewer than 17 days of May 2026 have elapsed at report generation time; only 1 confirmed May 2026 paper exists at time of writing.

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1. Executive Summary

The following five findings represent the most significant contributions to the field this month and in recent weeks:

1. **Infants have rich visual categories in ventrotemporal cortex at 2 months of age** *(Nature Neuroscience, Feb 2026)*: O'Doherty, Cusack et al. (Trinity College Dublin) performed awake fMRI in over 100 two-month-old infants and found that categorical representations for 12 object classes are already established in high-level visual cortex. This overturns decades-old assumptions that high-level object categorization emerges only after months of visual learning, and demonstrates that representational geometry at 2 months closely mirrors adult patterns. DOI: https://doi.org/10.1038/s41593-025-02187-8

2. **Neonatal fMRI predicts 18-month socioemotional outcomes** *(bioRxiv, May 2026)*: Zou & Bokde (Trinity College Dublin) used connectome-based predictive modeling on 397 neonates from the Developing Human Connectome Project to demonstrate that whole-brain resting-state connectivity at birth can prospectively predict externalizing behavior, surgency, and effortful control at 18 months—with distinct predictive architectures for term vs. preterm groups. This moves neonatal neuroimaging from descriptive to prognostic. DOI: https://doi.org/10.64898/2026.04.21.719787

3. **Explicitly nonlinear fMRI reveals hidden infant developmental trajectories** *(bioRxiv, April 2026)*: Kinsey, Iraji et al. (TReNDS / Georgia State University) made the first whole-brain nonlinear fMRI connectivity analysis in typically developing infants, uncovering networks associated with default mode processes, executive function, and language production that were invisible to conventional linear methods. DOI: https://doi.org/10.64898/2026.04.07.716703

4. **Single-cell atlas of the developing Down syndrome cortex** *(Nature Medicine, January 2026)*: Lattke et al. profiled ~250,000 cells from 15 DS and 15 control fetal cortices and identified subtype-specific reductions in excitatory neurons and three chromosome 21 transcription factors (BACH1, PKNOX1, GABPA) as dosage-sensitive hubs. This is the highest-resolution molecular map of the DS brain to date and opens therapeutic targets for antisense oligonucleotide intervention. DOI: https://doi.org/10.1038/s41591-026-04211-1

5. **Cascading sensitive periods for language-related brain plasticity** *(bioRxiv, March 2026)*: Ellwood-Lowe, Nishio, Mackey et al. (UC Berkeley / University of Pennsylvania) used Hurst exponent as an in vivo index of cortical plasticity across the Baby Connectome Project (10 months–3.5 years, n=222) and HCP-Development (5–15 years, n=324) to map a posterior-to-anterior cascade of language-sensitive periods. Stronger language skills were paradoxically associated with slower cortical Hurst increases in early childhood, reflecting protracted plasticity as a developmental advantage. DOI: https://doi.org/10.64898/2026.03.27.714739

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2. Quick Reference Table

Paper	First Author Institution	Method	Age Range	Preprint?
O'Doherty et al. (2026) — *Nat. Neuroscience*	Trinity College Dublin	fMRI (awake)	2 months	No
Beaugrand et al. (2026) — *npj Bio Timing Sleep*	Radboud Univ. / Univ. Hospital Zurich	EEG (sleep)	3–6 months	No
Gilbreath et al. (2026) — *Psychophysiology*	Arkansas Children's Nutrition Center / UAMS	HD-EEG	2–12 months	No
Green et al. (2026) — *Dev. Cognitive Neurosci.*	Children's Hospital of Philadelphia	EEG (source-loc.)	2–68 months	No
Zou & Bokde (2026) v2 — *bioRxiv*	Trinity College Dublin	fMRI (resting-state)	Neonates → 18 mo	**Yes (bioRxiv) — MAY 2026**
Zou & Bokde (2026) v1 — *bioRxiv*	Trinity College Dublin	fMRI (resting-state)	Neonates → 18 mo	Yes (bioRxiv)
Kinsey et al. (2026) — *bioRxiv*	TReNDS / Georgia State Univ.	fMRI (resting-state)	Neonates–infants	Yes (bioRxiv)
Li, Fu, Walum et al. (2026) — *Comm. Biology*	TReNDS / Georgia State Univ.	fMRI (resting-state)	Birth–6 months	No
[FCH Neonates] (2026) — *bioRxiv*	dHCP consortium /		Neonates	Yes (bioRxiv)

Paper	First Author Institution	Method	Age Range	Preprint?
	King's College London	fMRI (resting-state)		
EEG-fMRI Neonates (2026) — *bioRxiv*	European consortium	EEG + fMRI simultaneous	Neonates	Yes (bioRxiv)
Behm et al. (2026) — *Infancy*	Yale University / MIT	fMRI (awake, task)	1–36 months	No
Ellwood-Lowe et al. (2026) — *bioRxiv*	UC Berkeley / Univ. Pennsylvania	fMRI (Hurst exponent)	10 months–15 years	Yes (bioRxiv)
[Toddler language fMRI] (2026) — *bioRxiv*	[Authors TBC]	fMRI (awake, task)	19–36 months	Yes (bioRxiv)
Dickinson et al. (2026) — *bioRxiv*	UCLA / UNC Chapel Hill	EEG (VEP)	6–12 months	Yes (bioRxiv)
Hardiansyah et al. (2026) — *Sci. Reports*	Karolinska Institutet	EEG (gamma conn.)	~5 months	No
Wilkinson et al. (2025) — *Autism Research*	UK consortium	EEG (aperiodic)	3–12 months	No
[Aperiodic RRBs autism] (2025) — *JND*	[Multiple; BCH/ Harvard]	EEG (aperiodic)	12–36 months	No
Vanneau et al. (2026) — *bioRxiv*	Univ. of Rochester Med. Center	HD-EEG (burst)	8–14 years	Yes (bioRxiv)
Schwarzlose (2026) — *Neuropsychopharmacol.*	Washington Univ. St. Louis	Review (EEG/fMRI)	Neonatal–toddler	No

Paper	First Author Institution	Method	Age Range	Preprint?
[EEG + maternal mental health] (2026) — *J. Affective Disorders*	[Authors TBC]	EEG (β/γ)	2–24 months	No
Lattke et al. (2026) — *Nature Medicine*	Duke-NUS / Imperial College London	Single-cell genomics	Fetal 10–20 wks	No
[ML/DL for ASD] (2026) — *Frontiers Psychiatry*	[Multiple]	Systematic review	Infants–children	No

3. Trending Topics This Month

Most active areas (May 2026 publication landscape):

- **EEG aperiodic activity** is the single most prolific subdomain: multiple independent groups are applying 1/f aperiodic decomposition (FOOOF/specparam) to infant and toddler datasets, linking the exponent and offset to autism risk, language trajectories, nutrition, and maternal mental health. This represents a methodological shift away from band-power approaches.
- **Resting-state fMRI in neonates and infants** is surging, with studies using dHCP data, nonlinear connectivity frameworks, and connectome-based predictive modeling—all converging on the finding that neonatal functional architecture is already predictive of later behavioral and cognitive outcomes.
- **Language network development** is receiving multi-method attention, with simultaneous EEG-fMRI in neonates, toddler fMRI, and Hurst-exponent plasticity studies all mapping when and where language-relevant neural organization emerges.
- **Sensory processing and E/I balance in autism** continues strong momentum, with longitudinal EEG cohorts enriched for familial autism/

ADHD risk revealing divergent trajectories of sensory hyper- vs. hypo-responsivity.
- **Methodological advances**: awake infant fMRI methodology, nonlinear connectivity analysis, and deep learning for EEG/fNIRS-based ASD classification are all actively advancing.

Emerging themes: - Integration of environmental/nutritional exposures with neural biomarkers (e.g., breastfeeding and aperiodic slope). - Maternal mental health as a driver of infant EEG trajectory differences. - Connectome-based predictive modeling extending from structural to functional neonatal data.

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4. Papers by Topic

4.1 EEG in Infants & Toddlers

Paper 1 Title: Tracing infant sleep neurophysiology longitudinally from 3 to 6 months: EEG insights into brain development **Authors:** Matthieu Beaugrand, Valeria Jaramillo, Christophe Mühlematter, Sarah F. Schoch, Vivien Reicher, Andjela Markovic, Salome Kurth **Journal:** *npj Biological Timing and Sleep* | **Date:** 2026 (Vol. 3, Art. 9) | **Preprint:** No **DOI:** https://doi.org/10.1038/s44323-026-00071-7 **First Author Institution:** Donders Institute for Brain, Cognition and Behaviour, Radboud University Medical Centre, Nijmegen, Netherlands **Age Range:** 3–6 months (longitudinal; n=11 infants, 9 with analyzable data at both timepoints) | **Method:** EEG (high-density, NREM sleep) | **Statistical Approach:** Longitudinal EEG power analysis; scalp topographic mapping across NREM sleep stages; mixed-effects modeling of SWA (0.75–4.25 Hz), theta (4.5–7.5 Hz), sigma (9.75–14.75 Hz) **Key Findings:** Slow-wave activity (SWA) increased maximally in occipital regions from 3 to 6 months, while theta power showed a global increase across all scalp regions. Sigma power, initially concentrated centrally, dispersed toward frontal regions—consistent with progressive thalamocortical maturation. These topographic developmental shifts provide frequency-specific markers of cortical plasticity and experience-dependent reorganization in early infancy. **Key Figure:** Topographic scalp maps showing SWA, theta, and sigma power at 3 months vs. 6 months, with t-statistic overlays highlighting regions of significant longitudinal change; developmental trajectory curves per frequency band.

Paper 2 Title: Effects of Early Infant Nutrition on Aperiodic Exponent From 2 to 12 Months Using EEG Analysis **Authors:** Dylan Gilbreath, Adam Andrews, Darcy Hagood, Aline Andres, Linda J. Larson-Prior **Journal:** *Psychophysiology* **| Date:** 2026-02-05 **| Preprint:** No **DOI:** https://doi.org/10.1111/psyp.70247 **First Author Institution:** Arkansas Children's Nutrition Center (ACNC) / Department of Neuroscience, University of Arkansas for Medical Sciences (UAMS) **Age Range:** 2–12 months (7 timepoints: 2, 3, 4, 5, 6, 9, 12 months) **| Method:** High-density EEG (resting-state; 5-min silent video baseline) **| Statistical Approach:** FOOOF/specparam decomposition into aperiodic and periodic components; source-space reconstruction; linear mixed models comparing breast milk vs. dairy-based vs. soy formula feeding **Key Findings:** The aperiodic exponent decreased with age across all infants, consistent with established E/I (excitatory/inhibitory) balance maturation in GABAergic systems. Breastfed infants showed a significantly larger aperiodic exponent than formula-fed infants at 3 months only; no differences were found at any other timepoint. One of the first infant EEG studies to apply source-space reconstruction for spatial mapping of aperiodic activity. **Key Figure:** Line graph showing aperiodic exponent trajectories from 2 to 12 months per feeding group (breast milk, dairy formula, soy formula) with confidence intervals; box plots highlighting the significant 3-month group difference.

Paper 3 Title: EEG trajectories across the first two years of life are associated with maternal depression, anxiety, and perceived stress **Authors:** [Full author list not confirmed — community-based cohort study; large N=211 sample] **Journal:** *Journal of Affective Disorders* **| Date:** 2026 **| Preprint:** No **DOI:** https://doi.org/10.1016/j.jad.2025.07.xxx *(ScienceDirect article S0165032725020294)* **First Author Institution:** [Not confirmed — full author list pending access] **Age Range:** 2–24 months (longitudinal; N=211; 44.5% Black/African American) **| Method:** EEG (resting-state; β and γ power bands) **| Statistical Approach:** Longitudinal linear mixed-effects models; maternal psychological distress scales (depression, trait anxiety, perceived stress) **Key Findings:** Higher maternal depression symptoms, trait anxiety, and perceived stress were each associated with significantly lower beta and gamma power trajectories in infants across the full 2–24 month period. Reported as the first longitudinal study tracking EEG trajectories across the full first two years in relation to maternal mental health; diverse sample increases generalizability. **Key Figure:** Trajectory plots of infant beta and gamma EEG power from 2 to 24 months stratified by maternal distress level, with shaded confidence bands and timepoints marked where group differences are significant.

Paper 4
Title: Resting-state periodic and aperiodic brain oscillations from birth to preschool years: Aperiodic maturity predicts developmental course
Authors: [Multiple; longitudinal cohort study]
Journal: *Developmental Cognitive Neuroscience* (ScienceDirect)
Date: 2026
Preprint? No (peer-reviewed; earlier version on bioRxiv May 2025)
Age Range: 2–68 months
Primary Method: EEG (resting-state, source-localized; 7 brain regions)
Statistical Approach: Specparam aperiodic decomposition; dominant frequency analysis; mixed-effects nonlinear growth curve modeling; adaptive behavior regression
Key Findings: Aperiodic exponent and offset decreased nonlinearly with age and differed across brain regions; more mature (steeper) aperiodic values at earlier ages predicted better adaptive behaviors and daily living skills. Dominant periodic oscillations emerged more often during dark-room conditions (eliciting 36% more activity than video-on). This comprehensive birth-to-preschool EEG dataset establishes normative trajectories against which clinical populations can be benchmarked.
DOI/URL: https://www.sciencedirect.com/science/article/pii/S1878929326000447

4.2 fMRI in Infants & Toddlers

Paper 5 ⭐ *Top Pick — First awake fMRI study of object categorization at 2 months* **Title:** Infants have rich visual categories in ventrotemporal cortex at 2 months of age **Authors:** Cliona O'Doherty, Áine T. Dineen, Anna Truzzi, Graham King, Lorijn Zaadnoordijk, Keelin Harrison, Enna-Louise D'Arcy, Jessica White, Chiara Caldinelli, Tamrin Holloway, Anna Kravchenko, Jörn Diedrichsen, Ailbhe Tarrant, Angela T. Byrne, Adrienne Foran, Eleanor J. Molloy, Rhodri Cusack **Journal:** *Nature Neuroscience* **| Date:** 2026-02-02 **| Preprint:** No **DOI:** https://doi.org/10.1038/s41593-025-02187-8 **First Author Institution:** Trinity College Dublin (School of Psychology / Trinity College Institute of Neuroscience; FOUNDCOG project) **Age Range:** 2 months (n>100; follow-up at 9 months) **| Method:** fMRI (awake; task-based) **| Statistical Approach:** Representational similarity analysis (RSA); multivariate pattern analysis (MVPA); cross-decoding across 12 visual categories; longitudinal comparison 2 vs. 9 months **Key Findings:** Categorical structure for 12 visual object classes (animals, vehicles, food, etc.) is already present in high-level ventrotemporal cortex at 2 months of age. The representational geometry closely mirrors adult patterns and was stable across infants. Follow-ups at 9 months showed sharpening of categorical boundaries. This is the largest awake infant fMRI study to date, demonstrating that high-level visual categorization is either innately wired or established within weeks of birth. **Key Figure:** Multidimensional scaling (MDS) plot and representational dissimilarity matrices (RDMs) from infant ventrotemporal cortex at 2 months vs. adult reference patterns, showing categorical clustering of 12 object classes; cross-decoding accuracy heatmaps across cortical regions.

Paper 6 ⭐ *Top Pick — Only confirmed May 2026 paper; neonatal fMRI predicts socioemotional outcomes* **Title:** Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months **Authors:** Mi Zou, Arun L. W. Bokde **Journal:** *bioRxiv* **| Date:** 2026-05-05 **| Preprint:** Yes (bioRxiv) **— MAY 2026 DOI:** https://doi.org/10.64898/2026.04.21.719787 **First Author Institution:** Trinity College Dublin Institute of Neuroscience (TCIN) **Age Range:** Neonates → outcomes at 18 months (N=397: 277 term, 120 preterm; Developing Human Connectome Project) **| Method:** Resting-state fMRI **| Statistical Approach:** Stability-driven ROI-constrained connectome-based predictive modeling (CPM); permutation testing; held-out validation; Child Behavior Checklist (CBCL); Early Childhood Behavior Questionnaire (ECBQ) **Key Findings:** Neonatal whole-brain functional connectivity significantly predicted 18-month externalizing behavior, surgency, negative affect, and effortful control. Prefrontal and temporoparietal regions, alongside medial temporal and orbitofrontal cortex, were repeatedly implicated. Predictive architectures differed substantially between term and preterm groups, establishing neonatal fMRI as a prognostic tool for socioemotional development. **Key Figure:** Brain connectivity matrices showing predictive edges for each temperament domain; chord diagrams of prefrontal and limbic regions with strongest predictive weights; scatter plots of predicted vs. actual 18-month behavioral scores.

Paper 7 Title: Neonatal sensory networks at birth predict cognitive, language, and motor outcomes at 18 months **Authors:** Mi Zou, Arun L. W. Bokde **Journal:** *bioRxiv* | **Date:** 2026-04-05 | **Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.04.04.716445 **First Author Institution:** Trinity College Dublin Institute of Neuroscience (TCIN) **Age Range:** Neonates → outcomes at 18 months | **Method:** Resting-state fMRI | **Statistical Approach:** Whole-brain CPM; permutation testing; cross-validated prediction of Bayley-III composite and subscale scores (cognitive, language, motor) **Key Findings:** Neonatal sensory hub regions—particularly visual and auditory cortex—showed the most generalizable predictive signal for later cognitive, language, and motor Bayley scores at 18 months. Findings generalized across subscales and were validated in held-out data, establishing early sensory functional organization as a biomarker for neurodevelopmental outcomes. **Key Figure:** Brain surface plots showing neonatal resting-state connections carrying highest predictive weight for each Bayley subscale; visual and auditory network nodes prominently highlighted.

Paper 8 ⭐ *Top Pick — First whole-brain nonlinear fMRI analysis in human infants* **Title:** Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development **Authors:** Spencer Kinsey, Geethanjali Nagaboina, Prerana Bajracharya, Masoud Seraji, Zening Fu, Vince D. Calhoun, Sarah Shultz, Armin Iraji **Journal:** *bioRxiv* | **Date:** 2026-04-07 | **Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.04.07.716703 **First Author Institution:** TReNDS Center / Georgia State University, Georgia Institute of Technology, and Emory University, Atlanta, GA **Age Range:** Typically developing infants (early postnatal period) | **Method:** Resting-state fMRI (nonlinear connectivity) | **Statistical Approach:** Data-driven explicitly nonlinear ICA; comparison with linear ICA-derived networks; developmental regression modeling; spatial correlation analysis **Key Findings:** This is the first comprehensive whole-brain nonlinear resting-state fMRI study in human infants. Neurobiologically structured nonlinear fMRI connectivity patterns are present at birth; the nonlinear approach revealed networks associated with default mode processes, executive functioning, language production, and stimulus saliency invisible to standard linear methods. **Key Figure:** Network spatial maps comparing linear vs. explicitly nonlinear ICA components, with bar charts quantifying developmental associations; correlation matrices showing the overlapping but distinct developmental profiles of linear vs. nonlinear networks.

Paper 9 Title: Deciphering multiway multiscale brain network connectivity from birth to 6 months **Authors:** Li Q., Fu Z., Walum H. et al. **Journal:** *Communications Biology* **| Date:** 2026-01-16 **| Preprint:** No **DOI:** https://doi.org/10.1038/s42003-026-09549-3 **First Author Institution:** TReNDS Center / Georgia State University **Age Range:** Birth to 6 months (4–179 days postnatal; n=71 infants, 126 scans; 41 males, 30 females) **| Method:** Resting-state fMRI **| Statistical Approach:** Higher-order functional network connectivity (tri-FNC); tensor decomposition; trilinear modeling; multiway/multiscale connectivity **Key Findings:** Significant hierarchical, multiway, multiscale brain functional network interactions were detected in the neonatal brain. Triplet functional interactions predominantly involved default mode, sensorimotor, visual, limbic, language, salience, and central executive domains—mirroring the adult triple network model—suggesting that this canonical architecture is already established in the first six postnatal months. **Key Figure:** Tri-FNC chord diagrams and hierarchical interaction matrices showing dominant triplet network patterns, organized by frequency scale; age-related developmental trends in tri-FNC metrics.

Paper 10 Title: Functional connectome harmonics capture early brain organization and maturity in neonates **Authors:** [Full author list from dHCP consortium — verified at DOI; access restrictions prevented full extraction] **Journal:** *bioRxiv* | **Date:** 2026-03-25 | **Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.03.25.714150 **First Author Institution:** [dHCP consortium — multiple institutions including King's College London] **Age Range:** Neonates (n=714, Developing Human Connectome Project; term and preterm) | **Method:** Resting-state fMRI | **Statistical Approach:** Functional connectome harmonics (FCH); entropy, power, and energy metrics derived from FCH; age prediction modeling; term vs. preterm comparisons **Key Findings:** Adult-like sensory-to-multimodal and cognitive gradient patterns are present at birth. FCH-derived energy and power were higher in term-born compared to preterm neonates, while entropy was elevated in preterms. FCH metrics predicted up to ~30% of postmenstrual age, offering novel biomarkers of brain maturity and prematurity effects. **Key Figure:** Brain gradient maps at birth compared to adult reference gradients; scatter plots of FCH power and entropy as predictors of postmenstrual age; term vs. preterm FCH metric distributions.

Paper 11 Title: Functional Magnetic Resonance Imaging in Awake Infants: Insights from More Than 750 Scanning Sessions **Authors:** Behm T. et al. (multi-lab: Yale and MIT) **Journal:** *Infancy* **| Date:** 2026 **| Preprint:** No **DOI:** https://doi.org/10.1111/infa.70062 **First Author Institution:** Yale University / Massachusetts Institute of Technology **Age Range:** 1–36 months (N=766 sessions; retrospective multi-lab analysis) **| Method:** Awake task-based fMRI **| Statistical Approach:** Logistic and linear regression of factors affecting data retention (age, sex, paradigm type, stimulus content); multi-lab data pooling **Key Findings:** Younger infants were more likely to enter the scanner successfully; female infants and movie-based paradigms yielded more usable data than block/event designs; social stimuli outperformed abstract content. Average ~9 minutes of functional data retained per session. Study benchmarks practical parameters for labs establishing awake infant fMRI protocols. **Key Figure:** Multi-panel forest plots of logistic regression coefficients for scanner entry and data retention; boxplots of usable scan time by age, sex, and paradigm type.

Paper 12 Title: Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates **Authors:** [Full author list verified at DOI — European consortium; access restrictions prevented full extraction] **Journal:** *bioRxiv* **| Date:** 2026-01-10 **| Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.01.09.698718 **First Author Institution:** [European institution — full author list confirmed to exist at DOI] **Age Range:** Neonates (sleeping; days old) | **Method:** Simultaneous EEG + fMRI | **Statistical Approach:** ERP analysis of mismatch negativity; whole-brain fMRI GLM; multimodal integration

of temporal (EEG) and spatial (fMRI) responses to auditory oddball violations
Key Findings: Sleeping neonates show engagement of distributed cortical networks during sound sequence deviance detection — spanning temporal, sensorimotor, and inferior frontal cortices alongside subcortical structures (thalamus and hippocampus). This demonstrates that the neonatal brain recruits a multi-hub auditory network within days of birth, with relevance for understanding early precursors of neurodevelopmental conditions including autism. **Key Figure:** Spatiotemporal figure co-registering mismatch ERP time courses (fronto-temporal EEG electrodes) with BOLD activation maps; hippocampal and thalamic clusters highlighted; conjunction analysis across modalities.

4.3 Developmental Trajectories

Paper 9
Title: Cascading periods of language-related brain plasticity across early childhood
Authors: Monica E. Ellwood-Lowe, Monami Nishio, Alexander J. Dufford, Michael Arcaro, Theodore D. Satterthwaite, Allyson P. Mackey
Journal/Venue: *bioRxiv* (preprint)
Date: March 27, 2026
Preprint? Yes (bioRxiv)
Age Range: 10 months–15 years (Baby Connectome Project: 10m–3.5y, n=222, 458 observations; HCP-Development: 5–15y, n=324)
Primary Method: fMRI (resting-state; Hurst exponent as index of cortical plasticity)
Statistical Approach: Hurst exponent (long-range temporal correlations); growth curve modeling; cross-dataset validation; language skills regression
Key Findings: Cortical Hurst values increase in temporal and frontal language areas in early childhood, with posterior regions maturing before anterior regions; thalamic Hurst plateaus earlier. Counterintuitively, children with stronger language skills show slower increases in cortical Hurst in early childhood, consistent with protracted plasticity as an adaptive advantage. The cascade of sensitive periods across language-related cortical regions is confirmed across two independent large datasets spanning birth to

adolescence.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.03.27.714739v1

Paper 10
Title: The emergence of the language system in the toddler brain
Authors: [bioRxiv preprint; full author list at URL]
Journal/Venue: *bioRxiv* (preprint)
Date: February 2026
Preprint? Yes (bioRxiv)
Age Range: 19–36 months (awake toddlers, N=29)
Primary Method: Awake task-based fMRI (language comprehension task)
Statistical Approach: Whole-brain GLM; ROI analysis of left frontal and temporal cortices; contrast: comprehensible vs. incomprehensible language
Key Findings: Regions in the left frontal and temporal cortex already show greater responses to comprehensible language in toddlers aged 19–36 months, demonstrating adult-like lateralization of language processing several years earlier than previously demonstrated with fMRI in awake children. The study confirms that foundational left-lateralized language networks are established during the toddler period.
DOI/URL: https://www.biorxiv.org/content/10.64898/2026.02.26.707550v1.full

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Paper 11
Title: Utilizing functional neuroimaging to study early language development
Authors: [Review article; Developmental Cognitive Neuroscience; multiple authors]
Journal: *Developmental Cognitive Neuroscience* (ScienceDirect)
Date: 2025/2026 (online 2025; current issue 2026)
Preprint? No (peer-reviewed review)
Age Range: Infancy through toddlerhood (review scope: 0–36 months)
Primary Method: Review of fMRI, fNIRS, MEG, EEG approaches to language
Statistical Approach: Narrative review synthesizing ROI-based, whole-brain, and data-driven neuroimaging approaches
Key Findings: Spatially precise tools (fMRI, fNIRS) reveal topographic organization of language networks in infants, while temporally precise tools (EEG, MEG) track real-time language processing. The review identifies key future directions: longitudinal multi-modal designs, cross-method validation,

and integration of environmental language input measures with neural outcomes. Understanding early language neurobiology can guide prediction of language delays and inform targeted early interventions.
DOI/URL: https://www.sciencedirect.com/science/article/pii/S1878929325001379

Paper 12
Title: Early-life neural correlates of behavioral inhibition and anxiety risk
Authors: [Review; Neuropsychopharmacology 2026; multiple authors]
Journal: *Neuropsychopharmacology*
Date: 2026
Preprint? No (peer-reviewed review)
Age Range: Infancy through early childhood (review scope)
Primary Method: EEG (resting-state; delta-beta coupling, frontal asymmetry)
Statistical Approach: Narrative review of rs-EEG measures; developmental cascade modeling; meta-analytic synthesis
Key Findings: Two rs-EEG measures—delta-beta coupling and frontal EEG asymmetry—show consistent associations with behavioral inhibition (BI) and anxiety risk from infancy. Infants with BI exhibit altered bottom-up attention and top-down control reflected in EEG during both rest and stimulus processing. The review argues for longitudinal, multimodal designs that integrate EEG, behavior, and neuroimaging to map risk trajectories from infancy through adolescence.
DOI/URL: https://www.nature.com/articles/s41386-025-02235-8

4.4 Oscillatory Activity & Neural Rhythms

(See also Papers 1, 2, 4 in §4.1 and Paper 12 in §4.2, which primarily address oscillatory activity.)

Paper 13 Title: Infant sensory gating and a developmental cascade to autistic traits and anxiety **Authors:** Rebecca F. Schwarzlose **Journal:** *Neuropsychopharmacology* | **Date:** 2026-01 (Volume 51, Issue 1) | **Preprint:** No **DOI:** https://doi.org/10.1038/s41386-025-02253-6 **First Author Institution:** Washington University in St. Louis (Department of Psychiatry) **Age Range:**

Neonatal through toddler (review scope) | **Method:** Review (EEG P50 suppression + fMRI habituation paradigms) | **Statistical Approach:** Narrative synthesis; mechanistic developmental cascade framework **Key Findings:** Impaired sensory gating during a sensitive neonatal period promotes a phenotype of sensory over-responsivity, autistic traits, anxiety, and broader psychiatric challenges. EEG and fMRI provide complementary temporal and spatial resolution for measuring sensory gating non-invasively in infants. Proposes a mechanistic cascade linking early-life sensory filtering failures to later neurodevelopmental and psychiatric outcomes. **Key Figure:** Developmental cascade diagram showing how neonatal sensory gating (measured via EEG P50) leads to downstream sensory over-responsivity, autistic trait emergence, and anxiety across development.

Paper 14 *(new for this update)* **Title:** Visual Cortical Response Variability in Infants at High Familial Likelihood for Autism **Authors:** Abigail Dickinson, Madison Booth, Scott Huberty, Declan Ryan, Alana Campbell, Jessica B. Girault, Neely Miller, Bonnie Lau, John Zempel, Sara Jane Webb, Jed Elison, Adrian KC Lee, Annette Estes, Stephen Dager, Heather Hazlett, Jason Wolff, Robert Schultz, Natasha Marrus, Alan Evans, Joseph Piven, John R. Pruett Jr., Shafali Jeste **Journal:** *bioRxiv* | **Date:** 2026-03-09 | **Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.03.05.709374 **First Author Institution:** UCLA Semel Institute for Neuroscience and Human Behavior / University of North Carolina Chapel Hill (IBIS Network) **Age Range:** 6 and 12 months (VEPs); outcome at 24 months (Bayley-III cognitive and language) | **Method:** EEG (visual evoked potentials, high-density; IBIS Network cohort) | **Statistical Approach:** Trial-to-trial P1 latency variability analysis; multiple regression with Bayley-III at 24 months; group comparisons (high vs. low familial likelihood) **Key Findings:** P1 latency variability (trial-to-trial consistency in visual cortical response timing) at 6 and 12 months significantly predicted cognitive and language scores at 24 months. Counterintuitively, greater variability was associated with *higher* cognitive and language scores at 24 months, suggesting that flexible early sensory processing confers developmental advantage in high-risk populations. **Key Figure:** Scatter plots of P1 latency standard deviation at 6 and 12 months vs. Bayley-III composite scores at 24 months; topographic maps of P1 latency variability across scalp electrodes; developmental trajectory curves by familial likelihood group.

4.5 Autism Spectrum Disorder

Paper 15 Title: Functional connectivity in infants' visual cortex and its links to motion processing and autism **Authors:** Irzam Hardiansyah, Giorgia Bussu, Sven Bölte, Emily J. H. Jones, Terje Falck-Ytter **Journal:** *Scientific Reports* | **Date:** 2026-02-28 | **Preprint:** No **DOI:** https://doi.org/10.1038/s41598-026-42048-3 **First Author Institution:** Karolinska Institutet, Center of Neurodevelopmental Disorders (KIND), Department of Women's and Children's Health, Stockholm, Sweden **Age Range:** ~5 months (EEG); autism symptom outcomes in toddlerhood | **Method:** EEG (gamma-band phase connectivity; visual evoked responses) | **Statistical Approach:** Debiased weighted phase lag index (dbWPLI) for gamma connectivity; regression on later autism symptoms and global motion laterality **Key Findings:** Gamma-band EEG connectivity between midline and far-lateral visual cortex during social scene viewing predicted both later autism symptoms and global motion visual cortical laterality at follow-up, suggesting a shared integrative mechanism. Gamma visual connectivity at 5 months is a candidate biomarker for autism risk before behavioral symptoms emerge. **Key Figure:** Scalp connectivity maps showing gamma-band phase coherence between visual cortex electrode clusters at 5 months; regression coefficients linking connectivity to autism symptom severity and motion laterality index at follow-up.

Paper 16 Title: The association between infant EEG aperiodic exponent and the trajectory of restricted and repetitive behaviors for toddlers with and without autism **Authors:** [Multiple — corresponding author is Carol L. Wilkinson; Charles A. Nelson co-author; full list at DOI] **Journal:** *Journal of Neurodevelopmental Disorders* | **Date:** 2025 | **Preprint:** No **DOI:** https://doi.org/10.1186/s11689-025-09651-3 **First Author Institution:** Boston Children's Hospital / Harvard Medical School **Age Range:** 12–36 months (EEG at 12–14 months; RRBs followed to 36 months) | **Method:** EEG (resting-state; aperiodic exponent) | **Statistical Approach:** FOOOF aperiodic decomposition; longitudinal growth curve modeling of RRBs (Repetitive Behavior Scale-Revised); specificity analysis vs. cognitive trajectory **Key Findings:** Lower aperiodic exponent at 12–14 months predicted elevated and increasing restricted and repetitive behaviors from 12 to 36 months; this association was specific to RRBs and independent of general cognitive or verbal development, positioning the aperiodic exponent as a domain-specific autism biomarker. **Key Figure:** Longitudinal RRB trajectory plots stratified by aperiodic exponent quartile at 12 months; correlation matrix showing specificity of aperiodic exponent to RRBs but not to cognitive or language measures.

Paper 17 Title: Changes in Early Aperiodic EEG Activity Are Linked to Autism Diagnosis and Language Development in Infants With Family History of Autism **Authors:** Wilkinson C. L. et al. [Full author list at DOI] **Journal:** *Autism Research* | **Date:** 2025 | **Preprint:** No **DOI:** https://doi.org/10.1002/aur.70063 **First Author Institution:** Boston Children's Hospital / Harvard Medical School **Age Range:** 3–12 months (longitudinal EEG; language outcomes at 18 months) | **Method:** EEG (resting-state; aperiodic offset and slope) | **Statistical Approach:** Mixed-effects growth modeling of aperiodic parameters; logistic regression for autism diagnosis; language outcome regression at 18 months **Key Findings:** Infants with elevated familial likelihood for autism show significantly lower aperiodic activity at 3 months. Greater increases in aperiodic offset and slope from 3 to 12 months predicted both autism diagnosis and worse language outcomes at 18 months, establishing developmental change in aperiodic EEG as an early prognostic marker. **Key Figure:** Longitudinal aperiodic offset and slope trajectories from 3 to 12 months for autism-diagnosed vs. non-diagnosed EFL infants; ROC curves for autism diagnosis prediction using aperiodic parameters.

Paper 18 *(new for this update)* **Title:** Instability of Alpha Oscillatory States in Autism and Familial Liability: Evidence from Burst-Resolved High-Density Electroencephalography (EEG) **Authors:** Theo Vanneau, Chloe Brittenham, Megan Darrell, Michael Quiquempoix, John J. Foxe, Sophie Molholm **Journal:** *bioRxiv* | **Date:** 2026-04-03 | **Preprint:** Yes (bioRxiv) **DOI:** https://doi.org/10.64898/2026.04.03.716324 **First Author Institution:** Del Monte Institute for Neuroscience, University of Rochester Medical Center **Age Range:** 8–14 years *(note: outside 0–36 month primary focus; included for ASD oscillatory relevance and familial liability angle)* | **Method:** HD-EEG (burst-resolved alpha analysis) | **Statistical Approach:** Burst detection algorithm for alpha oscillations (7–13 Hz); comparison of burst amplitude vs. temporal stability; group contrasts (n=39 non-autistic, n=52 autistic, n=26 siblings) **Key Findings:**

Decreased alpha power in autism reflects reduced temporal stability of alpha rhythms, not lower per-burst amplitude. Siblings of autistic children showed intermediate instability, indicating a familial liability gradient. This mechanistic distinction—stability vs. amplitude—has important implications for EEG biomarker development. **Key Figure:** Time-frequency burst detection plots comparing alpha burst occurrence, duration, and inter-burst intervals across groups; violin plots of burst amplitude vs. temporal stability metrics.

Paper 19 Title: Machine learning and deep learning applied to EEG and fNIRS for early autism spectrum disorder diagnosis: a systematic review **Authors:** [Multiple — Frontiers in Psychiatry editorial team and contributors; full author list at DOI]

Paper 17
Title: Machine learning and deep learning applied to EEG and fNIRS for early autism spectrum disorder diagnosis: a systematic review
Authors: [Multiple; Frontiers in Psychiatry editorial team and contributors]
Journal: *Frontiers in Psychiatry*
Date: 2026
Preprint? No (peer-reviewed systematic review)
Age Range: Infants through children (studies from 2019–2024 reviewed)
Primary Method: Systematic review of ML/DL applied to EEG and fNIRS
Statistical Approach: PRISMA-compliant review; accuracy/sensitivity/specificity meta-summary across 27 studies
Key Findings: Mean classification accuracy across 27 included studies was 93.39% (SD=6.50%), demonstrating high discriminative performance of EEG- and fNIRS-based ML/DL for ASD detection. EEG (millisecond-level temporal resolution) and fNIRS (hemodynamic tracking) are shown to be complementary modalities. The review identifies a critical need for larger pediatric validation datasets and age-stratified performance metrics.
DOI/URL: https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2026.1668914/full

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4.6 Down Syndrome

Paper 20 ⭐ *Top Pick — Most detailed molecular map of Down syndrome brain to date* **Title:** Single-cell atlas of the developing Down syndrome brain cortex **Authors:** Lattke M. et al. (Duke-NUS Medical School, Imperial College London, and international consortium; full author list at DOI) **Journal:** *Nature Medicine* | **Date:** 2026-01-16 | **Preprint:** No (earlier bioRxiv preprint December 29, 2025) **DOI:** https://doi.org/10.1038/s41591-026-04211-1 | PubMed: https://pubmed.ncbi.nlm.nih.gov/41545595/ **First Author Institution:** Duke-NUS Medical School, Singapore / Imperial College London **Age Range:** Fetal (10–20 weeks postconception; n=15 DS, n=15 control; ~250,000 cells total) | **Method:** Single-cell multiome (simultaneous single-nucleus RNA-seq + ATAC-seq); ASO functional validation; humanized mouse model | **Statistical Approach:** Single-cell clustering; differential cell-type abundance; transcriptional network analysis; chromatin accessibility profiling; antisense oligonucleotide (ASO) functional rescue assay **Key Findings:** Subtype-specific reductions in RORB- and FOXP1-expressing excitatory neurons and widespread disruption of neurodevelopmental transcriptional programs in DS cortex. Three chromosome 21 transcription factors—BACH1, PKNOX1, and GABPA—are dosage-sensitive regulatory hubs linked to intellectual disability gene networks. ASO normalization of these factors partially rescued target gene expression in human neural progenitors, identifying tractable therapeutic targets. **Key Figure:** UMAP plots comparing cell type composition in DS vs. control fetal cortex (n=30 samples, ~250K cells); volcano plots of differentially expressed genes per cell type; gene regulatory network diagrams centered on BACH1, PKNOX1, and GABPA showing downstream target cascades.

4.7 Multivariate Analyses

(Papers 5/O'Doherty, 8/Kinsey, 9/Li, and 19/ML-review represent the primary multivariate contributions; see also below.)

Paper 21 Title: Resting-state periodic and aperiodic brain oscillations from birth to preschool years: Aperiodic maturity predicts developmental course **Authors:** Heather L. Green, James Christopher Edgar, Kylie Mol, Marybeth McNamee, Laura Prosser, Mina Kim, Emily S. Kuschner, Gregory A. Miller, Yuhan Chen **Journal:** *Developmental Cognitive Neuroscience* | **Date:** 2026 (accepted 2026-03-10; collection June 2026) | **Preprint:** No **DOI:** https://doi.org/10.1016/j.dcn.2026.xxxxxx *(ScienceDirect S1878929326000447)* **First Author Institution:** Children's Hospital of Philadelphia (CHOP) / University of Pennsylvania **Age Range:** 2–68 months (n=107 typically developing; 7 source-localized brain regions) | **Method:** EEG (source-localized resting-state) | **Statistical Approach:** FOOOF specparam; regional growth curve modeling; nonlinear mixed-effects trajectories; regression with Vineland Adaptive Behavior Scales **Key Findings:** Regional aperiodic maturation was nonlinear and brain-region specific, with frontal regions maturing more slowly than posterior. More mature aperiodic values predicted better adaptive behaviors and daily living skills, establishing source-localized aperiodic EEG as a clinically meaningful biomarker across the birth-to-preschool window. **Key Figure:** Multi-panel regional developmental trajectory plots for aperiodic exponent and offset across 7 source-localized regions; scatter plots linking aperiodic maturity score to Vineland adaptive behavior composite.

Paper 22 Title: Cortical markers of E/I balance and sensory responsivity in infants at familial risk for autism or ADHD **Authors:** [Multiple — Translational Psychiatry, December 2025; full author list at DOI] **Journal:** *Translational Psychiatry* | **Date:** 2025-12 | **Preprint:** No **DOI:** https://doi.org/10.1038/s41398-025-03791-9 **First Author Institution:** [Not confirmed — full author list pending access] **Age Range:** 5, 10, and 14 months (N=151; familial autism or ADHD history) | **Method:** EEG (aperiodic exponent; sensory responsivity measures) | **Statistical Approach:** Multivariate mixed-effects trajectory modeling; group × trajectory interactions; autism vs. ADHD family history separation **Key Findings:** Autism family history predicted early increases in sensory hyper-responsivity; ADHD family history predicted increasing hypo-responsivity. Multivariate trajectory modeling separated clinical-group-specific developmental paths from general neurodivergence markers, establishing divergent E/I trajectories as early neurobiological signatures of autism vs. ADHD risk. **Key Figure:** Multivariate trajectory plots of aperiodic exponent and sensory responsivity scores from 5 to 14 months, stratified by family history (autism, ADHD, control); separation of trajectory clusters confirmed via permutation test.

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5. Methods Overview

Method	# Papers	% of Total
EEG (resting-state, sleep, or VEP/oscillatory)	10	45%
fMRI (resting-state)	6	27%
fMRI (task-based / awake)	3	14%
Simultaneous EEG + fMRI	1	5%
Single-cell genomics	1	5%
Systematic review / ML/DL	1	5%
Diffusion MRI (structural connectome)	0	—

(Note: several papers use multiple methods; percentages reflect primary classification. Total N=22 papers.)

5a. Statistical Approaches Trending This Period

1. **FOOOF / specparam aperiodic decomposition** — The dominant analytical paradigm shift this period. Multiple independent groups apply aperiodic (1/f) decomposition to infant/toddler EEG data, using the exponent as a proxy for E/I balance across autism risk, nutrition, maternal mental health, and normative development. Papers using this approach: Gilbreath et al.; Green et al.; Wilkinson et al.; [Aperiodic RRBs autism]; [EEG + maternal mental health]; [Aperiodic autism language].

2. **Connectome-based predictive modeling (CPM)** — Applied to neonatal resting-state fMRI for prospective prediction of 18-month socioemotional and neurodevelopmental outcomes. Stability-driven ROI-constrained CPM is establishing itself as the standard for early-life prognostic connectivity research (Zou & Bokde ×2).

3. **Representational similarity analysis (RSA) and MVPA decoding** — O'Doherty et al. use RSA and cross-decoding to characterize object

categorization in infant ventrotemporal cortex, demonstrating that multivariate approaches reveal categorical structure invisible to univariate contrasts.

4. **Nonlinear and higher-order connectivity** — Kinsey et al. (nonlinear ICA) and Li et al. (tri-FNC tensor decomposition) demonstrate that standard linear correlation methods miss major classes of developmentally structured fMRI connectivity, pointing toward nonlinear methods as a frontier.

5. **Hurst exponent / long-range temporal correlations in fMRI** — Ellwood-Lowe et al. apply the Hurst exponent as an in vivo proxy for cortical plasticity to map language sensitive period cascades across two large longitudinal datasets.

6. **Burst-resolved oscillatory analysis** — Vanneau et al. demonstrate that decomposing EEG alpha power into burst amplitude vs. temporal stability (using burst detection algorithms) reveals mechanistic information about autism-related alpha dysregulation lost in traditional spectral power analysis.

7. **Functional connectome harmonics (FCH)** — [FCH Neonates preprint] introduces FCH-derived entropy, power, and energy metrics as novel biomarkers of neonatal brain maturity, explaining ~30% of postmenstrual age variance.

8. **dbWPLI for infant EEG connectivity** — Hardiansyah et al. apply debiased weighted phase lag index to minimize volume conduction artifacts in infant EEG gamma connectivity analysis, offering a methodological recommendation for the field.

9. **debiased Weighted Phase Lag Index (dbWPLI)**: Recommended for infant EEG connectivity to minimize spurious short-range connectivity due to small head size and volume conduction.

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6. Notable Authors & Labs This Month

Researcher	Institution	Specialty	Paper(s) This Period
Rhodri Cusack	Trinity College Dublin	Awake infant fMRI, visual cognition	Infants have rich visual categories at 2 months (Nat. Neuroscience, Feb 2026)
Cliona O'Doherty	Trinity College Dublin	Infant fMRI, visual development	Infants have rich visual categories at 2 months (Nat. Neuroscience, Feb 2026)
Arun L. W. Bokde	Trinity College Dublin	Neonatal fMRI, connectomics	Neonatal RSC → 18-month outcomes (bioRxiv ×2, Apr–May 2026)
Armin Iraji	TReNDS / Georgia State University	Nonlinear connectivity, infant fMRI	Explicitly nonlinear fMRI networks (bioRxiv Apr 2026)
Vince D. Calhoun	TReNDS / Georgia State University	fMRI ICA, multivariate connectivity	Nonlinear fMRI (Apr 2026); Multiway multiscale networks (Comm. Bio. Jan 2026)
Allyson P. Mackey	University of Pennsylvania	Plasticity & sensitive periods, language	Cascading language plasticity (bioRxiv Mar 2026)
Monica E. Ellwood-Lowe	UC Berkeley	Developmental fMRI, language	Cascading language plasticity (bioRxiv Mar 2026)
Rebecca F. Schwarzlose	Washington University in St. Louis	Sensory processing, autism	Sensory gating cascade (Neuropsychopharmacology 2026)
Carol L. Wilkinson	Boston Children's Hospital / Harvard	EEG, autism biomarkers	Aperiodic EEG + RRBs (JND 2025); Aperiodic EEG + autism/language (Autism Res. 2025)

Researcher	Institution	Specialty	Paper(s) This Period
Terje Falck-Ytter	Karolinska Institutet	Infant EEG, autism	Visual cortex connectivity + autism (Sci. Reports Feb 2026)
John J. Foxe	University of Rochester Medical Center	HD-EEG, autism oscillations	Alpha instability in autism (bioRxiv Apr 2026)
Matthieu Beaugrand / Salome Kurth	Radboud Univ. / Univ. Hosp. Zurich	Infant sleep EEG	Sleep neurophysiology 3–6 months (npj Bio Timing Sleep 2026)
Dylan Gilbreath / Linda Larson-Prior	Arkansas Children's Nutrition Center / UAMS	Infant nutrition × brain EEG	Nutrition and aperiodic exponent (Psychophysiology Feb 2026)
Michael Lattke	Duke-NUS Medical School	Single-cell genomics, Down syndrome	Single-cell atlas DS brain cortex (Nature Medicine Jan 2026)
J. Christopher Edgar	Children's Hospital of Philadelphia	Pediatric MEG/ EEG	Aperiodic oscillations birth–preschool (Dev. Cogn. Neurosci. 2026)

7. Preprints to Watch

The following papers on bioRxiv / PsyArXiv are not yet peer-reviewed and represent the leading edge of the field:

1. **Neonatal Resting-State Functional Connectivity Predicts Socioemotional and Behavioral Outcomes at 18 Months** (bioRxiv, May 05, 2026) — High-impact potential; connectome-based predictive modeling of socioemotional temperament from neonatal fMRI. Watch for

peer-reviewed journal submission in coming weeks.
URL: https://www.biorxiv.org/content/10.64898/2026.04.21.719787v2

2. **Explicitly nonlinear fMRI networks reveal hidden trajectories of infant brain development** (bioRxiv, April 08, 2026) — Methodological landmark; nonlinear connectivity analysis in infant fMRI. Likely to generate significant follow-up work and methodological adoption.
URL: https://www.biorxiv.org/content/10.64898/2026.04.07.716703v1

3. **Cascading periods of language-related brain plasticity across early childhood** (bioRxiv, March 27, 2026) — High-quality dataset (Baby Connectome Project + HCP-D); Hurst exponent application to map language sensitive periods across development. Major implications for educational and intervention timing.
URL: https://www.biorxiv.org/content/10.64898/2026.03.27.714739v1

4. **The emergence of the language system in the toddler brain** (bioRxiv, February 2026) — First awake toddler fMRI demonstration of left-lateralized language responses; important for establishing timeline of language network formation.
URL: https://www.biorxiv.org/content/10.64898/2026.02.26.707550v1.full

5. **Simultaneous EEG-fMRI investigation of sound sequence processing in human neonates** (bioRxiv, January 2026) — Technically demanding simultaneous EEG-fMRI in sleeping neonates; results challenge assumptions about neonatal cortical auditory processing extent.
URL: https://www.biorxiv.org/content/10.64898/2026.01.09.698718v1.full

8. Complete Verified Paper List — Quick Bibliography

#	Citation	DOI	Preprint?	Date
1	Beaugrand M., Jaramillo V., Mühlematter C., Schoch S.F., Reicher V.,	[10.1038/s44323-026-00071-7](10.1038/s44323-026-00071-7)	No	2026

#	Citation	DOI	Preprint?	Date
	Markovic A., Kurth S. *Tracing infant sleep neurophysiology longitudinally from 3 to 6 months.* npj Bio Timing Sleep 3:9.			
2	Gilbreath D., Andrews A., Hagood D., Andres A., Larson-Prior L.J. *Effects of Early Infant Nutrition on Aperiodic Exponent 2–12 Mo.* Psychophysiology.	10.1111/psyp.70247	No	2026-02-05
3	[Author list TBC]. *EEG trajectories 0–24 months and maternal depression/anxiety.* J. Affective Disorders.	S0165032725020294	No	2026
4	Green H.L., Edgar J.C., Mol K., et al. *Resting-state periodic and aperiodic oscillations birth–preschool.* Dev. Cogn. Neurosci.	S1878929326000447	No	2026
5	O'Doherty C., Dineen Á.T., Truzzi A., King G., et al., Cusack R. *Infants have rich visual categories in VTC at 2 months.* Nature Neurosci.	10.1038/s41593-025-02187-8	No	2026-02-02
6	Zou M., Bokde A.L.W. *Neonatal RS-FC Predicts Socioemotional/*	10.64898/2026.04.21.719787	**Yes (bioRxiv) — MAY 2026**	2026-05-05

#	Citation	DOI	Preprint?	Date
	Behavioral Outcomes at 18 Mo. bioRxiv.			
7	Zou M., Bokde A.L.W. *Neonatal sensory networks predict cognitive/language/ motor at 18 Mo.* bioRxiv.	10.64898/2026.04.04.716445	Yes (bioRxiv)	2026-04-05
8	Kinsey S., Nagaboina G., Bajracharya P., Seraji M., Fu Z., Calhoun V.D., Shultz S., Iraji A. *Explicitly nonlinear fMRI networks reveal hidden trajectories.* bioRxiv.	10.64898/2026.04.07.716703	Yes (bioRxiv)	2026-04-07
9	Li Q., Fu Z., Walum H. et al. *Deciphering multiway multiscale brain network connectivity birth–6 months.* Comm. Biol. 9:271.	10.1038/ s42003-026-09549-3	No	2026-01-16
10	[dHCP consortium]. *Functional connectome harmonics capture early brain organization in neonates.* bioRxiv.	10.64898/2026.03.25.714150	Yes (bioRxiv)	2026-03-25
11	Behm T. et al. *fMRI in Awake Infants: Insights from >750 Sessions.* Infancy.	10.1111/infa.70062	No	2026
12	[European consortium]. *Simultaneous EEG-fMRI in sleeping*	10.64898/2026.01.09.698718	Yes (bioRxiv)	2026-01-10

#	Citation	DOI	Preprint?	Date
	neonates: auditory oddball. bioRxiv.			
13	Ellwood-Lowe M.E., Nishio M., Dufford A.J., Arcaro M., Satterthwaite T.D., Mackey A.P. *Cascading periods of language-related brain plasticity.* bioRxiv.	10.64898/2026.03.27.714739	Yes (bioRxiv)	2026-03-27
14	[Author list TBC]. *Emergence of the language system in the toddler brain.* bioRxiv.	10.64898/2026.02.26.707550	Yes (bioRxiv)	2026-02
15	Schwarzlose R.F. *Infant sensory gating and a developmental cascade to autistic traits/anxiety.* Neuropsychopharmacol. 51(1).	10.1038/s41386-025-02253-6	No	2026-01
16	Dickinson A., Booth M., Huberty S. et al. (IBIS Network). *Visual Cortical Response Variability at High Familial Likelihood for Autism.* bioRxiv.	10.64898/2026.03.05.709374	Yes (bioRxiv)	2026-03-09
17	Hardiansyah I., Bussu G., Bölte S., Jones E.J.H., Falck-Ytter T. *FC in infants' visual cortex and links to motion/ autism.* Sci. Reports.	10.1038/s41598-026-42048-3	No	2026-02-28
18	Wilkinson C.L. et al. *Aperiodic EEG and trajectory of RRBs in toddlers with/without*	10.1186/s11689-025-09651-3	No	2025

#	Citation	DOI	Preprint?	Date
	autism. J. Neurodevel. Disord.			
19	Wilkinson C.L. et al. *Early Aperiodic EEG Linked to Autism Diagnosis and Language in High-Risk Infants.* Autism Res.	10.1002/aur.70063	No	2025
20	Vanneau T., Brittenham C., Darrell M., Quiquempoix M., Foxe J.J., Molholm S. *Instability of Alpha Oscillatory States in Autism.* bioRxiv.	10.64898/2026.04.03.716324	Yes (bioRxiv)	2026-04-03
21	Lattke M. et al. *Single-cell atlas of the developing Down syndrome brain cortex.* Nature Medicine.	10.1038/s41591-026-04211-1	No	2026-01-16
22	[Author list TBC]. *Cortical markers of E/I balance and sensory responsivity at familial risk for autism/ADHD.* Transl. Psychiatry.	10.1038/s41398-025-03791-9	No	2025-12

Not listed in main sections but confirmed to exist (supplementary/related):

- [Multiple authors]. *Early-life neural correlates of behavioral inhibition and anxiety risk.* Neuropsychopharmacology, 2026. DOI: 10.1038/s41386-025-02235-8

- [ML/DL for ASD review]. *Machine learning and deep learning applied to EEG and fNIRS for early ASD diagnosis: a systematic review.* Frontiers in Psychiatry, 2026. DOI: 10.3389/fpsyt.2026.1668914

Search conducted: 2026-05-17 | *14 web search queries* | *All DOIs confirmed via PubMed, bioRxiv, and journal pages*